8 Storing data on hard drives: EBS and instance store

This chapter covers

- Attaching persistent storage volumes to an EC2 instance
- Using temporary storage attached to the host system
- Backing up volumes
- Testing and tweaking volume performance
- Differences between persistent (EBS) and temporary volumes (instance store)

Imagine your task is to migrate an enterprise application being hosted on-premises to AWS. Typically, legacy applications read and write files from a filesystem. Switching to object storage, as described in the previous chapter, is not always possible or easy. Fortunately, AWS offers good old block-level storage as well, allowing you to migrate your legacy application without the need for expensive modifications.

Block-level storage with a disk filesystem (FAT32, NTFS, ext3, ext4, XFS, and so on) can be used to store files as you would on a personal computer. A *block* is a sequence of bytes and the smallest addressable unit. The OS is the intermediary between the application that needs to access files and the underlying filesystem and block-level storage. The disk filesystem manages where (at what block address) your files are stored. You can use block-level storage only in combination with an EC2 instance where the OS is running.